Yihang Qin

Shanghai Jiao Tong University, Shanghai, China

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Education

M.Phil., Shanghai Jiao Tong University **2024.09 – Present**
Relevant Course: Artificial Neural Network, Graphs and Networks, Data Mining *Shanghai, China*

B.Eng., Harbin Institute of Technology (Shenzhen) **2020.09 – 2024.06**
Relevant Course: Advanced Mathematics, Automatic Control, Language Programming *Shenzhen, China*

Research Interest
Cooperation, Game Theory, Reinforcement Learning, Human-AI Interaction

Publications

Yihang Qin & L. Wang. (2026). Exploring cooperation mechanisms via reinforcement learning in network common-pool resource games. *arXiv preprint arXiv:2606.05867*. arXiv:2606.05867.

Yihang Qin, H. Chen, & L. Wang. (2026). Recovering cooperation in Spatial Prisoner's Dilemma via a forced loner mechanism. *Physica A: Statistical Mechanics and its Applications*, 131309. doi: 10.1016/j.physa.2026.131309.

Yihang Qin, H. Chen, & L. Wang. (2026). A tripartite intervention–conflict game with Frequency-Adjusted Q-learning. *Applied Mathematical Modelling*, 117072. doi: 10.1016/j.apm.2026.117072.

Yihang Qin, H. Chen, & L. Wang. (2025, July). An Tripartite Evolutionary Game Analysis of the 'Siege Wei to Rescue Zhao' Strategy. In *2025 44th Chinese Control Conference (CCC)* (pp. 8458–8463). IEEE. doi: 10.23919/CCC64809.2025.11179046.

Selected Research Projects

Exploring cooperation mechanisms via reinforcement learning in network common-pool resource games **2026.01 – Present**
First Author | *arXiv preprint*

- Established a networked common-pool resource games with resource allocation and cooperation dynamics.
- Developed a reinforcement learning-based agent responsible for allocation mechanisms to improve long-term resource sustainability and fairness.
- Used three baseline allocation mechanisms to fit and interpret the RL agent's policy, achieving better results.

A tripartite intervention–conflict game with Frequency-Adjusted Q-learning **2025.09 – 2026.05**
First Author | *Accepted by Applied Mathematical Modelling*

- Developed a tripartite evolutionary game model to analyze strategic interactions among interveners and conflicting parties.
- Established error bounds linking Frequency-Adjusted Q-learning and evolutionary game dynamics.
- Analyzed the effects of key parameters on system equilibria and stability.

Recovering cooperation in spatial prisoner's dilemma via a forced loner mechanism **2025.11 – 2026.01**
First Author | *Accepted by Physica A: Statistical Mechanics and its Applications*

- Proposed a resource-constraint dynamic model in spatial prisoner's dilemma games.

- Designed a forced loner mechanism inspired by social welfare safety nets to model resource-depletion-induced strategy transitions.
- Conducted numerical simulations to study cooperation level, resource sustainability, and spatial pattern formation.

Design of transmodal associative learning circuit based on memristor **2023.12 – 2024.05**

Bachelor's Thesis | *Harbin Institute of Technology, Shenzhen*

- Designed a memristor-based associative memory circuit inspired by neuromorphic learning and cross-modal association.
- Constructed storage neuron, synapse, and retrieval neuron modules based on an AIST memristor model.
- Implemented bimodal and multimodal associative learning circuits and verified learning, forgetting, recovery, transfer, and consolidation behaviors through PSPICE simulations.

Experience

"Zhuoyi Cup" · 2025 Second National Swarm Intelligence Challenge **2025.08 – 2025.11**

Core Developer | *First Prize in the Aerial Competition Track*

- Designed reinforcement learning-based strategies for multi-agent confrontation tasks.
- Built a rule-based expert system to collect demonstration data and used behavior cloning to warm-start policy learning.
- Contributed to code implementation, experimental evaluation, and the final competition presentation.

Research on Reinforcement Learning with Role Assignment Across Scales **2025.12 – 2026.05**

Undergraduate Research Mentor | *Undergraduate Thesis Project*

- Mentored an undergraduate thesis project on "Intelligent Game Modeling Based on Hierarchical Reinforcement Learning".
- Guided the setup of the SMAC-based StarCraft II multi-agent environment, as well as experimental design, result analysis, and thesis writing.

Nonlinear Systems Course **Spring 2025 & Spring 2026**

Teaching Assistant | *Shanghai Jiao Tong University*

- Served as a teaching assistant for the undergraduate course "Nonlinear Systems" over two spring semesters.
- Supported students by answering course-related questions and explaining key concepts, analytical methods, and problem-solving techniques.
- Responsible for grading homework assignments and exams.

Technical Skills

Research Methods: Evolutionary game theory, reinforcement learning, multi-agent systems, complex networks, numerical simulation.
Programming and Tools: Python, MATLAB, LaTeX, PyTorch, Git, Linux, PSPICE.
Languages: Mandarin Chinese (native), English (CET-6: 512; IELTS Academic: 7.0).
Interests: Reading, Music, Fitness, Hiking, Traveling, and Video games.